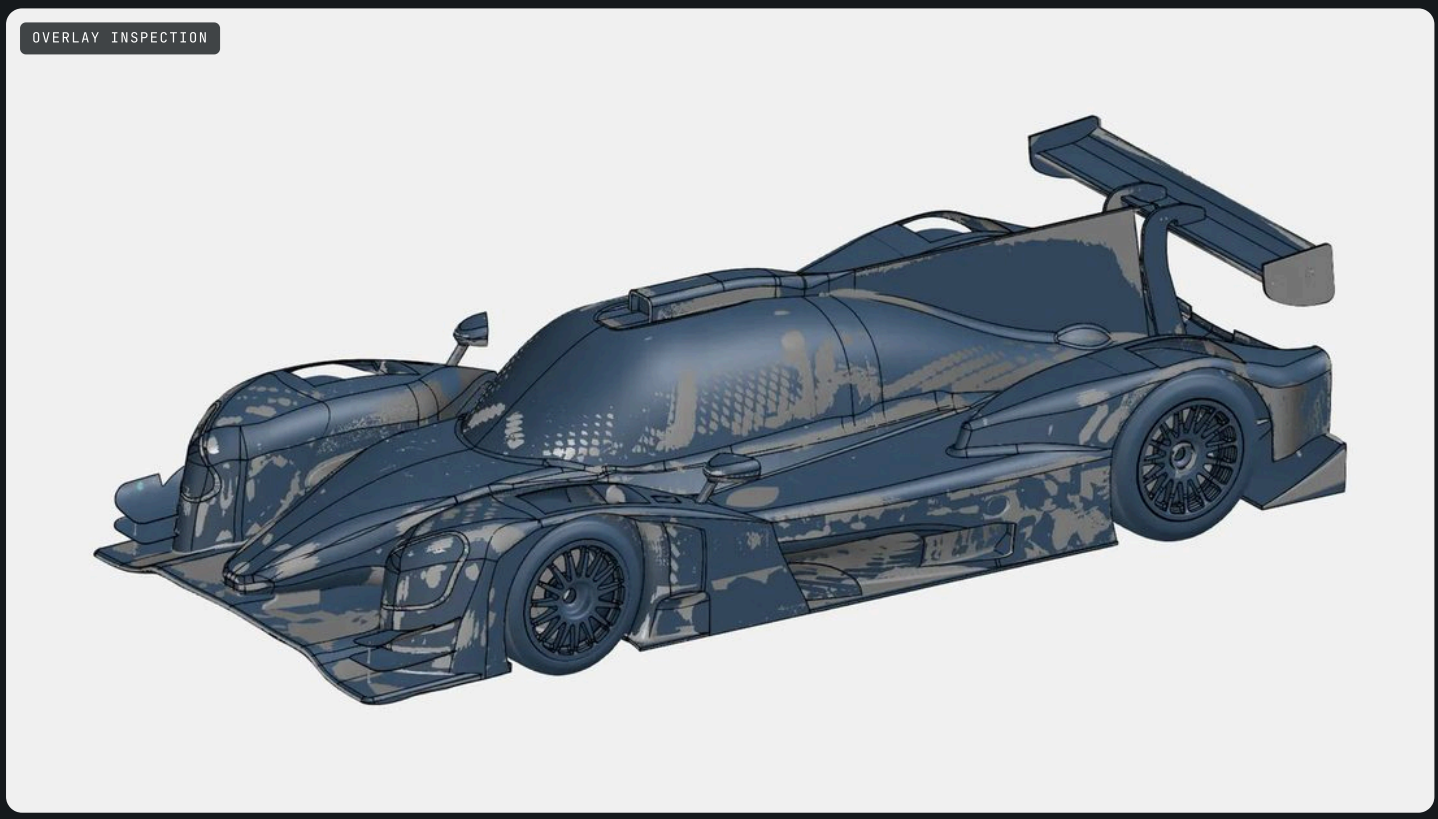
PORTFOLIO

3D Scanning & Inspection

Manufactured parts scanned and compared with their CAD, so you see exactly where a part is in or out of tolerance — fast, full-surface and without contact.



SHOWN: PROTOTYPE RACE CAR BODY

11	3	0.02 mm	IN · AE
PROJECTS IN THIS PORTFOLIO	INDUSTRIES	SCANNER PRECISION, UP TO	ON-SITE SCANNING

01 What we deliver

Compare manufactured parts to their CAD — to the micron.

We inspect manufactured parts by scanning them and comparing the result against the nominal CAD. Colour deviation maps, GD&T checks and first-article reports show exactly where a part is in or out of tolerance — fast, and without contact.

Colour deviation maps

Full-surface comparison against CAD, instantly showing where material is over or under.

GD&T inspection

Flatness, position, profile and more, measured and reported against the drawing.

First-article inspection

Complete FAI reports for new parts, tools and suppliers.

Trend analysis

Repeated inspections track tool wear and process drift over time.

02 Deliverables

- Inspection report (.PDF)
- Colour deviation maps
- GD&T results tables
- Scan data for records

03 How we work

Scan to solution in four steps.

STEP 01

Capture

We scan the part, product or site — in our lab or on location.

STEP 02

Process

Scans are aligned, cleaned and merged into usable, watertight data.

STEP 03

Engineer

CAD, Class A surfaces, BIM or inspection built from that data.

STEP 04

Deliver

CAD files, drawings and reports in the formats your team uses.

04 Industries in this portfolio

- Automobile 8
- Consumer Products 2
- Mining 1

05 Common questions

Is 3D scanning accurate enough to replace a CMM?

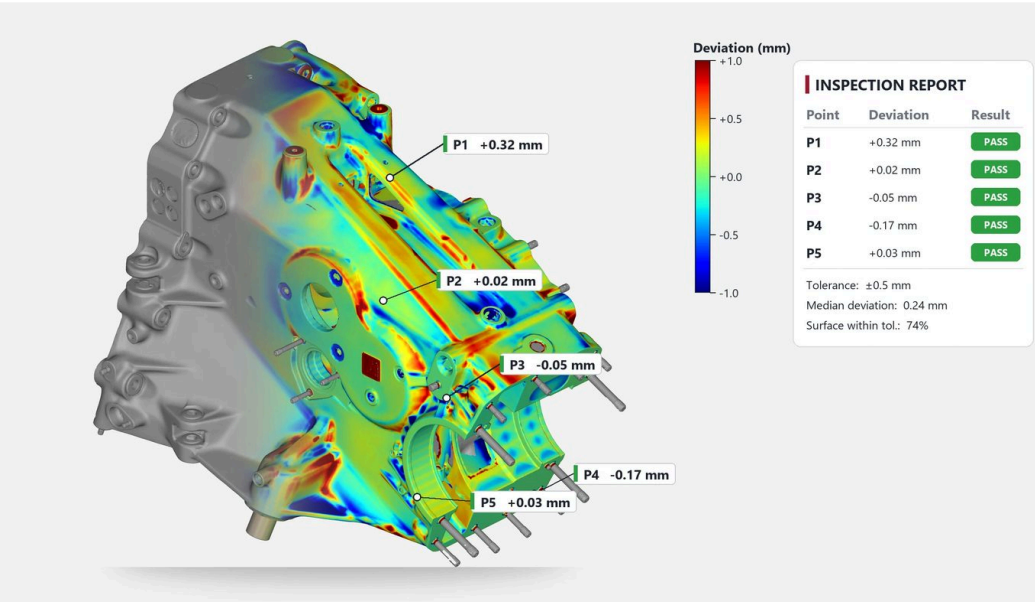
For most castings, moulded parts and freeform surfaces, yes. For micron-level machined features, a CMM is still the better tool, and many companies combine both.

What do you need from us for an inspection?

The part (or access to it on site), the CAD model, and the drawing or tolerances to check against. If there is no CAD, we can compare two parts against each other, or against a "golden" sample.

How long does a 3D inspection take?

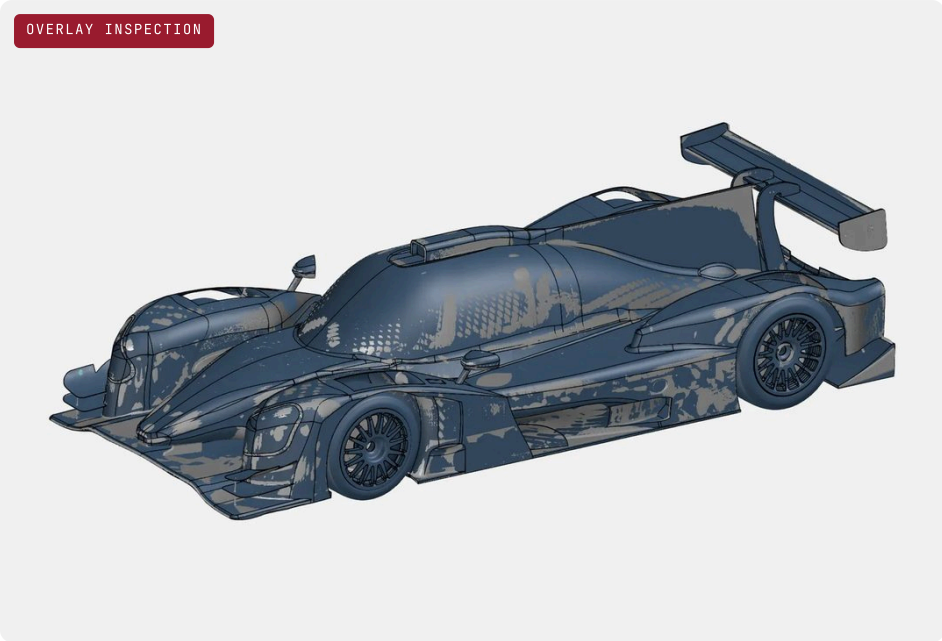
Scanning a typical part takes minutes to an hour. Reporting depends on how many features and GD&T checks you need.

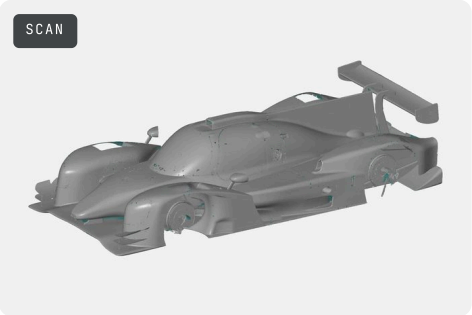


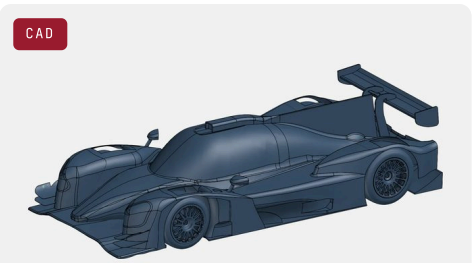
Real parts scanned and engineered by the VShark team.

01 Prototype Race Car Body

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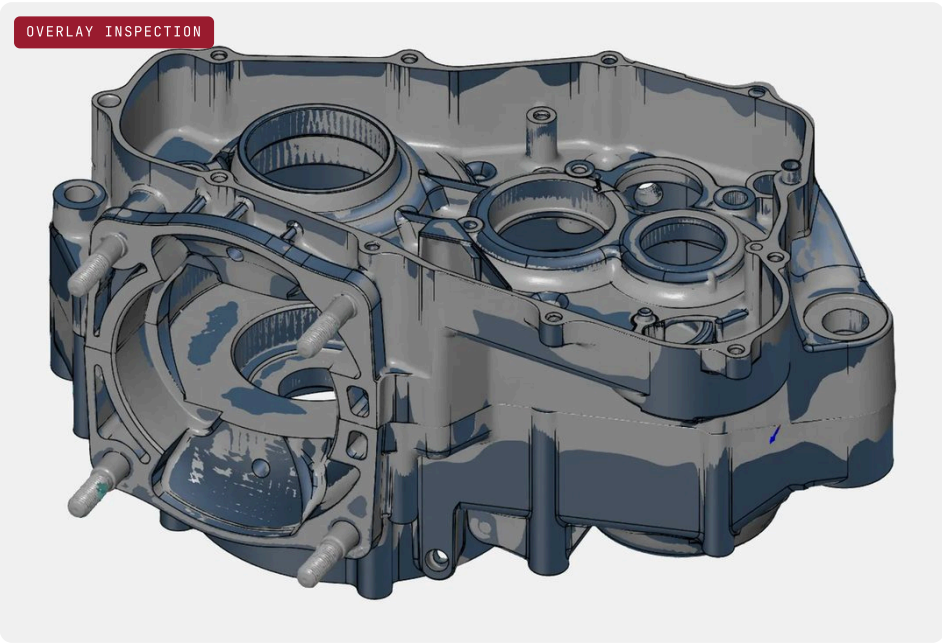


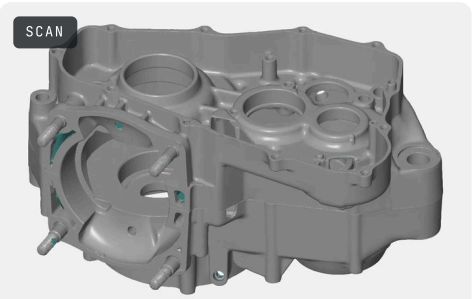


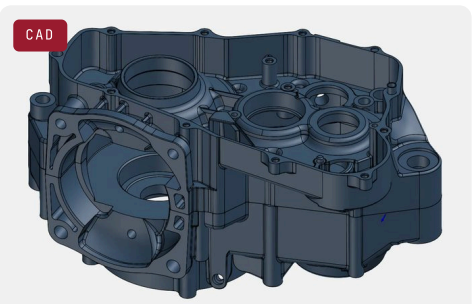
Full body of a prototype race car scanned, rebuilt as CAD and checked against the scan.

02 Engine Crankcase Casting

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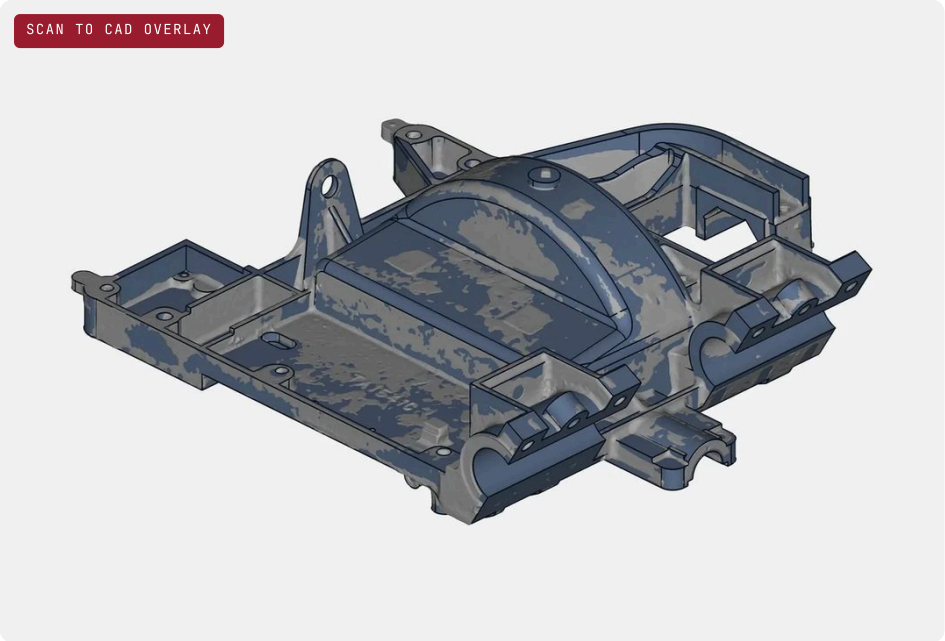


Complex crankcase casting digitised and remodelled, with a deviation overlay.

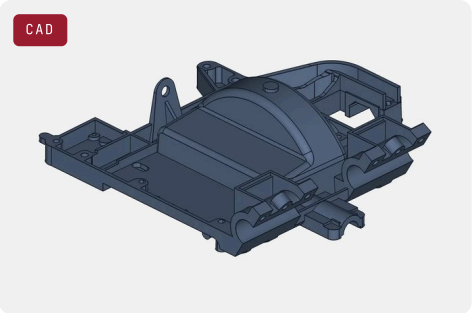
Real parts scanned and engineered by the VShark team.

03 Transmission Lower Housing Casting

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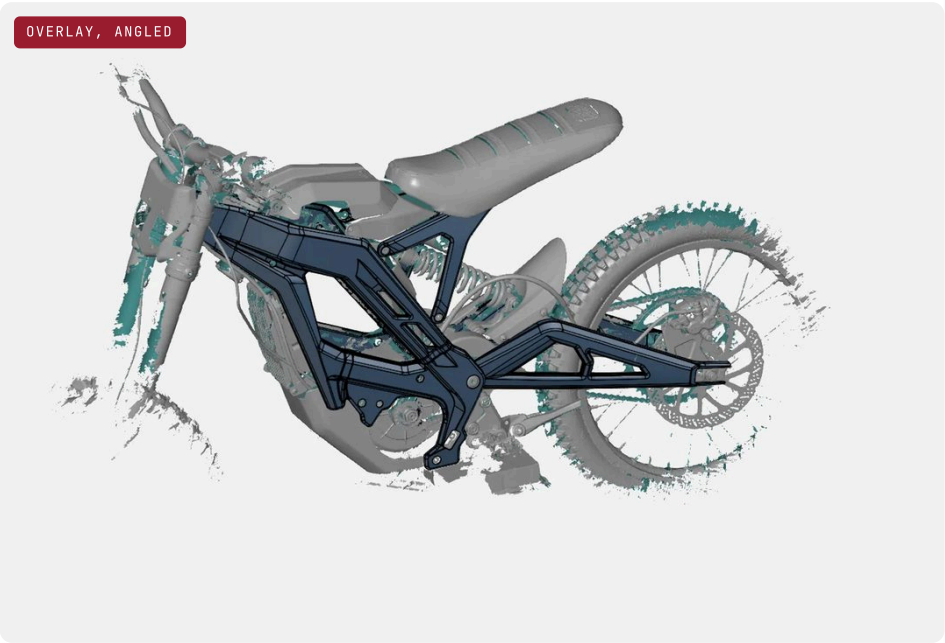


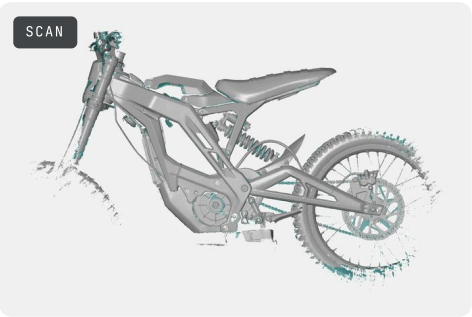


Transmission housing casting scanned, modelled and compared against the CAD.

04 Electric Dirt Bike Frame

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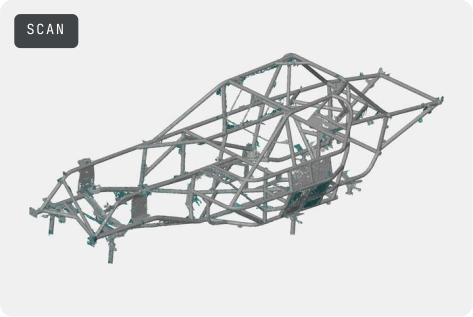
Whole bike scanned; the frame extracted and rebuilt as editable CAD.

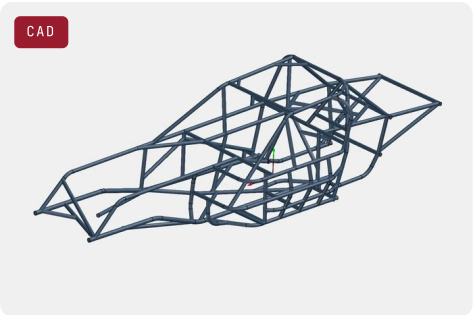
Real parts scanned and engineered by the VShark team.

05 Tubular Roll Cage

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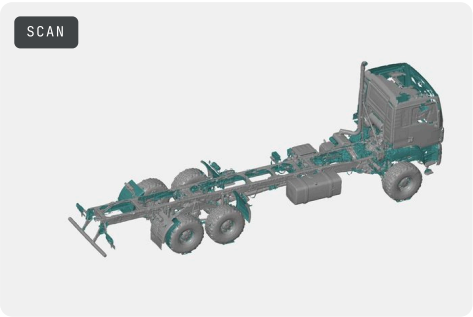


Space-frame roll cage captured tube by tube and modelled for fabrication.

06 Truck Chassis Frame

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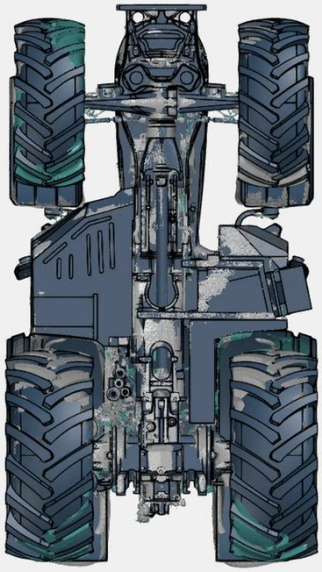
Heavy truck chassis scanned and rebuilt as a CAD assembly.

Real parts scanned and engineered by the VShark team.

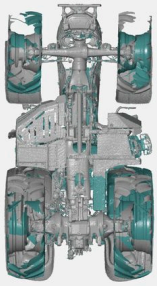
07 Tractor Underbody

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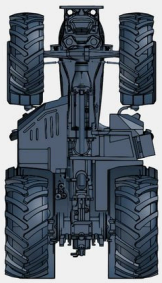
OVERLAY INSPECTION



SCAN



CAD

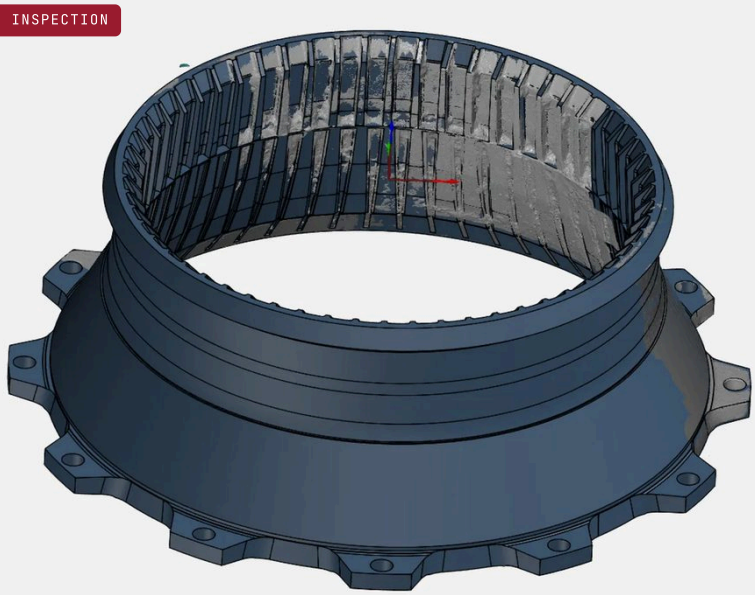


Tractor underbody digitised for redesign and packaging studies.

08 Cone Crusher Liner

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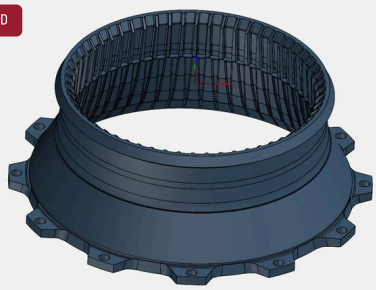
OVERLAY INSPECTION



SCAN



CAD



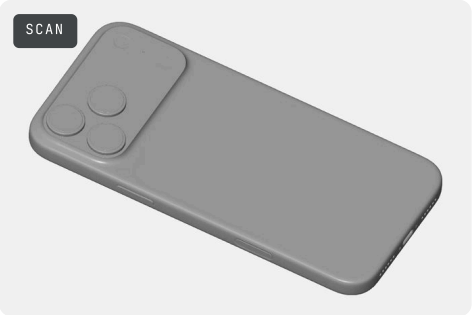
Wear liner scanned and modelled so replacements can be produced.

Real parts scanned and engineered by the VShark team.

09 Smartphone Back Housing

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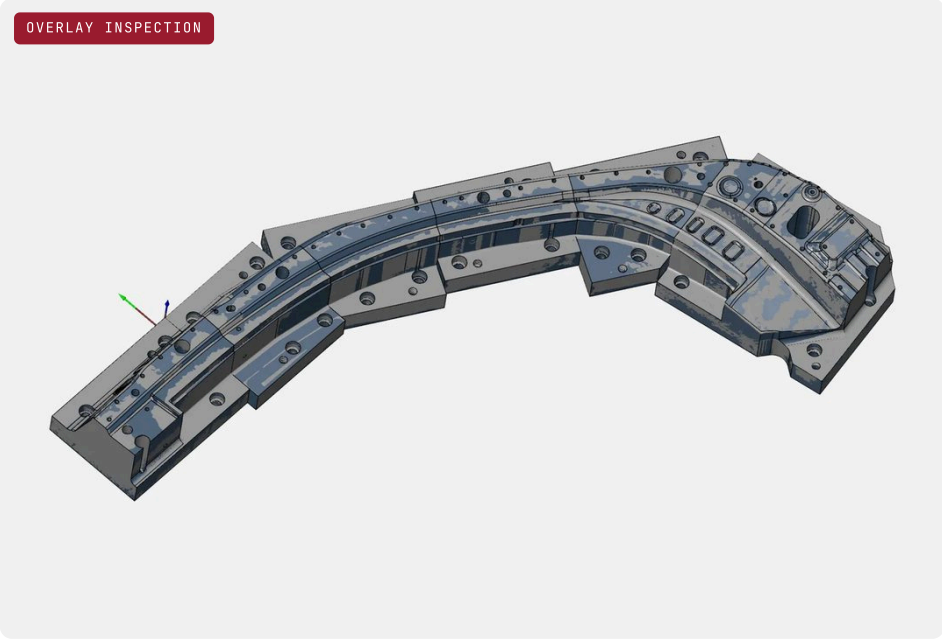


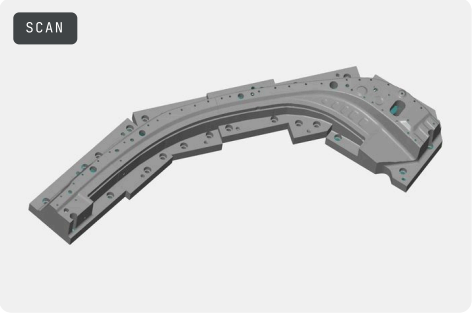


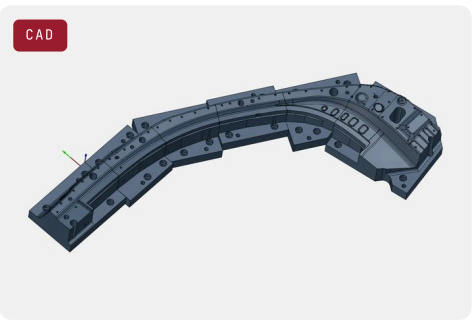
Thin-walled consumer housing captured and remodelled to design intent.

10 Automotive Mold Insert

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Mold insert assembly reverse-engineered for tool repair and duplication.

Real parts scanned and engineered by the VShark team.

11 Plastic Shopping Trolley

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OVERLAY INSPECTION



SCAN



CAD



Large moulded trolley scanned and rebuilt as production CAD.

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Need a part inspected?

Send the part (or give us access to it on site), the CAD model, and the drawing or tolerances to check against. If there is no CAD, we can compare two parts against each other, or against a "golden" sample.

We usually reply within one working day. NDAs on request — your files are used only for your project.

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